

Homework 5

EE 999 — Introduction to Control Systems

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Due Wednesday, October 14, 2026

Problem 1.A — Transfer function

$$U(s) = k_p R(s) \tag{1}$$

$$Y(s) = \frac{b_1}{s + a_1} U(s) \tag{2}$$

$$Y(s) = \frac{k_p b_1}{s + a_1} R(s) \tag{3}$$

$$G_{ry1}(s) = \frac{Y(s)}{R(s)} = \frac{k_p b_1}{s + a_1} \tag{4}$$

Problem 1.B — Transfer function

$$Y(s) = \frac{b_1}{s + a_1} U(s) \tag{5}$$

$$U(s) = \frac{k_p s + k_i}{s} R(s) \tag{6}$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \tag{7}$$

$$G_{ry2}(s) = \frac{Y(s)}{R(s)} = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} \tag{8}$$